

ADVANCE DATABASES

EXPERIMENT:01

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BCA/B4

EXPERIMENT:1.1

CODE:

```
1 |
2 DECLARE
3     A NUMBER := &A;
4     B NUMBER := &B;
5     C NUMBER := &C;
6
7 BEGIN
8     IF (A > B AND A > C) THEN
9         DBMS_OUTPUT.PUT_LINE('A is greatest: ' || A);
10    ELSIF (B > A AND B > C) THEN
11        DBMS_OUTPUT.PUT_LINE('B is greatest: ' || B);
12    ELSIF (C > A AND C > B) THEN
13        DBMS_OUTPUT.PUT_LINE('C is greatest: ' || C);
14    ELSE
15        DBMS_OUTPUT.PUT_LINE('Two or more values are equal / greatest');
16    END IF;
17 END;
```

OUTPUT:

Substitution Variables

A

3

Cancel

OK

Substitution Variables

B

Enter a value

Cancel

OK

Substitution

Substitution Variables

C

9

Substitution Variables

Cancel

OK

SQL> DECLARE

 A NUMBER := 3;

 B NUMBER := 8;

 C NUMBER := 9;...

Show more...

C is greatest: 9

PL/SQL procedure successfully completed.

Elapsed: 00:00:00.012

Experiment : 1

* Write a PL/SQL code to accept the value of A, B & C display which is greater.

DECLARE

A NUMBER := 10;

B NUMBER := 20;

C NUMBER := 30;

BEGIN

IF (A > B AND A > C) THEN

DBMS_OUTPUT.PUT_LINE ('A is Greater: ' || A);

ELSEIF (B > A AND B > C) THEN

DBMS_OUTPUT.PUT_LINE ('B is Greater: ' || B);

ELSEIF (C > A AND C > B) THEN

DBMS_OUTPUT.PUT_LINE ('C is Greater: ' || C);

ELSE

DBMS_OUTPUT.PUT_LINE ('Two or more values are equal & greatest');

END IF;

END;

/

EXPERIMENT:1.2

CODE:

```
DECLARE
    i NUMBER := 1;
BEGIN
    LOOP
        DBMS_OUTPUT.PUT_LINE('Welcome to PL/SQL Programming');
        i := i + 1;
        EXIT WHEN i > 5;
    END LOOP;
END;
/
```

OUTPUT:

```
Welcome to PL/SQL Programming  
Welcome to PL/SQL Programming  
Welcome to PL/SQL Programming  
Welcome to PL/SQL Programming  
Welcome to PL/SQL Programming
```

```
PL/SQL procedure successfully completed.
```

```
Elapsed: 00:00:00.007
```

Experiment: 1.2

Using PL/SQL statements create a simple loop that display message "Welcome to PL/SQL Programming" 5 times.

```
DECLARE
```

```
  i NUMBER := 1;
```

```
BEGIN
```

```
  LOOP
```

```
    DBMS_OUTPUT.PUT_LINE ("Welcome to PL/SQL  
                           Programming");
```

```
    i := i + 1;
```

```
    EXIT WHEN i > 5;
```

```
  END LOOP
```

```
END;
```

```
/
```


EXPERIMENT:1.3

CODE:

```
VARIABLE P NUMBER
EXEC : = 5;

DECLARE
    P NUMBER := 5;
    V_factorial NUMBER := 1;

BEGIN
    IF P < 0 THEN
        DBMS_OUTPUT.PUT_LINE('Factorial is not defined for negative numbers');
    ELSIF P = 0 THEN
        DBMS_OUTPUT.PUT_LINE('The factorial of 0 is: 1');
    ELSE
        FOR i IN 1 .. P LOOP
            V_factorial := V_factorial * i;
        END LOOP;
        DBMS_OUTPUT.PUT_LINE('The factorial of ' || P || ' is: ' || V_factorial);
    END IF;
END;
/
```



OUTPUT:

```
SQL> DECLARE
        P NUMBER := 5;
        V_factorial NUMBER := 1;
        ...
Show more...
```

The factorial of 5 is: 120

Experiment 1.3

Write a PL/SQL code block to find the factorial of number.

```
VARIABLE P_Number NUMBER
```

```
EXEC :P_number := 5;
```

```
DECLARE
```

```
  V_Factorial NUMBER := 1;
```

```
BEGIN
```

```
  IF :P_number < 0 THEN
```

```
    DBMS_OUTPUT.PUT_LINE ('factorial is not defined  
for negative numbers')
```

```
  ELSEIF :P_number = 0 THEN
```

```
    DBMS_OUTPUT.PUT_LINE ('The factorial of 0 is
```

```
  ELSE
```

```
    FOR i IN 1... P_number LOOP
```

```
      V_factorial := V_factorial * i;
```

```
    END LOOP;
```

```
    DBMS_OUTPUT.PUT_LINE ('The factorial of '|| P_number  
is '|| V_factorial || ');
```

```
  END IF;
```

```
END;
```

Teacher's Signature : _____

EXPERIMENT:1.4

CODE:

```
2
3 DECLARE
4     v_first_num    NUMBER := 0;
5     v_second_num   NUMBER := 1;
6     v_next_num     NUMBER;
7     v_limit        NUMBER := 10;
8 BEGIN
9     DBMS_OUTPUT.PUT_LINE('Fibonacci series up to ' || v_limit || ' terms:');
10
11     IF v_limit >= 1 THEN
12         DBMS_OUTPUT.PUT_LINE(v_first_num);
13     END IF;
14
15     IF v_limit >= 2 THEN
16         DBMS_OUTPUT.PUT_LINE(v_second_num);
17     END IF;
18
19     FOR i IN 3 .. v_limit LOOP
20         v_next_num := v_first_num + v_second_num;
21         DBMS_OUTPUT.PUT_LINE(v_next_num);
22         v_first_num := v_second_num;
23         v_second_num := v_next_num;
24     END LOOP;
25 END;
```

OUTPUT:

```
Fibonacci series up to 10 terms:
```

```
0
```

```
1
```

```
1
```

```
2
```

```
3
```

```
5
```

```
8
```

```
13
```

```
21
```

```
34
```

```
PL/SQL procedure successfully completed.
```

```
Elapsed: 00:00:00.003
```

1.4 Write a PL/SQL code to generate fibonacci series.

```
SET SERVEROUTPUT ON SIZE 1000000;
```

```
DECLARE
```

```
  V_first_num NUMBER := 0;
```

```
  V_second_num NUMBER := 1;
```

```
  V_next_num NUMBER;
```

```
  V_limit NUMBER := 10;
```

```
BEGIN
```

```
  DBMS_OUTPUT.PUT_LINE ('Fibonacci series up to  
    '||V_limit||' terms:');
```

```
  IF V_limit >= 1 THEN
```

```
    DBMS_OUTPUT.PUT_LINE (V_first_num);
```

```
  END IF;
```

```
  IF V_limit >= 2 THEN
```

```
    DBMS_OUTPUT.PUT_LINE (V_second_num);
```

```
  END IF;
```

```
  FOR i IN 3 .. V_limit LOOP
```

```
    V_next_num := V_first_num + V_second_num;
```

```
    DBMS_OUTPUT.PUT_LINE (V_next_num);
```

```
    V_first_num := V_second_num;
```

```
    V_second_num := V_next_num;
```

```
  END LOOP;
```

```
END;
```

Teacher's Signature : _____

EXPERIMENT:1.5

CODE:

```
DECLARE
    v_n    NUMBER := 10;
    v_sum  NUMBER := 0;
BEGIN
    FOR i IN 1 .. v_n LOOP
        v_sum := v_sum + i;
    END LOOP;

    DBMS_OUTPUT.PUT_LINE('The sum of first ' || v_n || ' numbers is: ' || v_sum);
END;
/
```

OUTPUT:

The sum of first 10 numbers is: 55

PL/SQL procedure successfully completed.

Elapsed: 00:00:00.007

7.5 Write a PL/SQL code to find sum of n no.

```
DECLARE
  Vn NUMBER := 10;
  Vsum NUMBER := 0;
BEGIN
  FOR i IN 1...Vn LOOP
    Vsum := Vsum + i;
  END LOOP;
  DBMS_OUTPUT.PUT_LINE ('The sum of first
    '||Vn||' number is: '||Vsum);
END;
```


